ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

CROP SCIENCE

6049/1

PAPER 1 Multiple Choice

NOVEMBER 2021 SESSION

1 hour

Additional materials:
Multiple Choice answer sheet,
Soft clean eraser,
Soft pencil (type B or HB is recommended),
Electronic calculator.

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.
Write your name, centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question there are **four** possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

Each correct answer will score one mark.

Any rough working should be done in this booklet.

This question paper consists of 10 printed pages and 2 blank pages.

[Turn over

1 At which stage of cell division does cytokinesis occur?

A metaphase
B anaphase
C telophase
D interphase

2 What is the best management option to reduce bulk density?

A intensive cultivation
B addition of crop residue
C continuous cropping
D deep ploughing

3 Which one is a role of boron on plant growth and development?

A nitrogen fixation
B chlorophyll formation
C growth of meristematic tissues
D conversion of nitrogen to protein

4 Which nutrient is responsible for reducing water movement and salt balance?

A calcium
B nitrogen
C phosphorus
D magnesium

5 Uneven seed germination is influenced by

A good soil-seed contact.
B poor imbibition.
C pressure potential.
D soil capping.

6 Which mechanism makes grains long lasting?

A exine
B intine
C pollen tube
D generative nucleus

7 ________ is the force used by plants to pull water from roots up to the leaves via the xylem.

A Root pressure
B Turgor pressure
C Osmotic pressure
D Transpirational pull

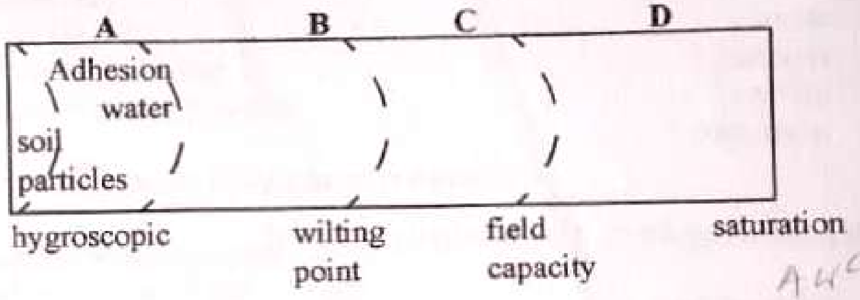
8 Pairing of homologous chromosomes is also known as

A synapsis.
B cytokinesis.
C chiasmata.
D crossing over.

9 Which one is a fungal disease?

A canker
B soft rot
C rosette
D powdery mildew

10 At which point is water available for plant growth on the diagram?



AWC = FC –

11 Which group consists of primary tillage implements only?

A plough, ripper, subsoiler
B plough, disc harrow, planter
C disc harrow, planter, cultivator
D cultivator, planter, ripper

12 The placement of seeds in a continuous row is termed

A dibbing.
B drill seeding.
C hill dropping.
D precision planting.

13 How much residue is left on the soil surface in stubble mulching?

A 10%
B 20%
C 30%
D 40%

14 Which soil requires more lime to raise its pH?

A clay soil with low organic matter
B sandy loam with high organic matter
C clay soil with high organic matter
D sandy soil with low organic matter

15 Which process breaks down a glucose molecule into two molecules of pyruvic acid?

A photolysis
B glycolysis
C hydrolysis
D phosphorylation

16 Which bud is induced by the shortening of days just before winter?

A shoot
B root bud
C axillary
D intercalary

17 Which growth regulator reduces lodging in plants?

A Gibberellins
B Chlormequat chloride
C Indole-acetic acid
D Tridobenzoic acid

18 Use of male sterility in plant breeding eliminates

A manual emasculation.
B self incompatibility.
C cross pollination.
D self pollination.

19 What is a wettable powder?

A Powders which are compatible in water.
B Pesticide in which the active ingredient is the solid phase.
C Active ingredient is formulated to act in the gaseous form.
D An active ingredient is formulated into a dry very fine powder for direct application.

20 Which factor influences insecticide efficiency?

A Type of insect
B Environmental conditions
C Adult stage of the insect
D Rate of application of the insecticides

21 Eukaryotic cells are

A complex organism without a nucleus.
B simpler structural organisms without a nucleus.
C ribosomes occupied extensively by DNA.
D characterised by numerous membrane bound organelles.

22 Mitotic divisions of the chromosomes without cytoplasmic division is called

A cytokinesis.
B karyokinesis.
C microsporogenesis.
D generative nucleogenesis.

23 A soil matrix is made of

A pore spaces only.
B solid particles only.
C mineral matter and water.
D solid particles and pore spaces.

24 A mature embryo sac in ovule development is formed by

A degeneration of embryo cells.
B repeated meiotic divisions.
C repeated mitotic divisions.
D nuclear fusion.

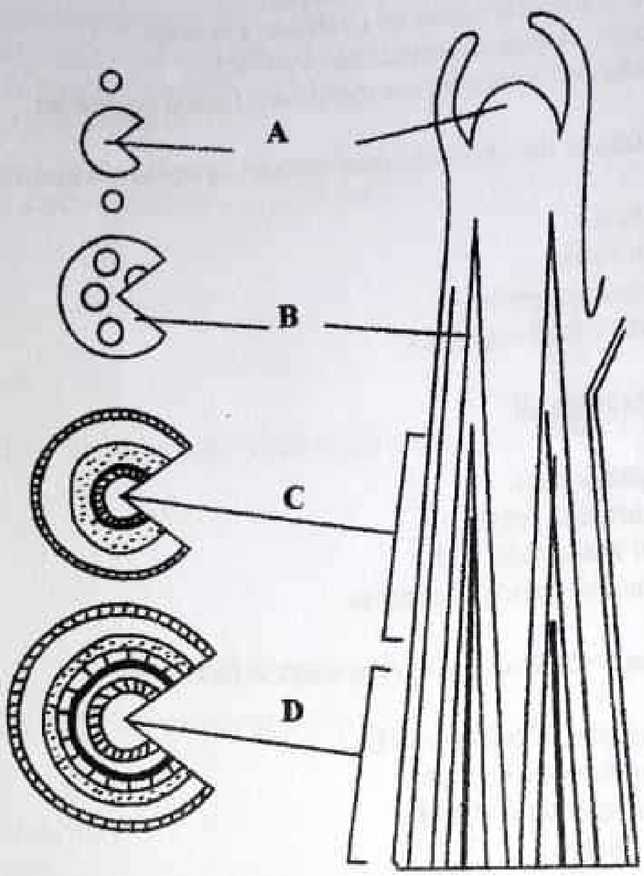
25 Using the Munsell system soil labelled **10YR3/6**. What does **10YR** stand for?

A 10 years of soil colour
B A yellow colour of soil
C A yellow red colour of soil
D 10 yellow and red soil colour

26 What causes soil colours of grey shades in waterlogged conditions?

A Iron oxides in their increased form
B Iron oxides in their stable form
C Iron oxides in their solid form
D Iron oxides in their reduced form

27 The diagram shows a shoot tip of a dicotyledon.



Which one is the zone of cell division?

28 The symplastic pathway explains that

A water moves through the cell wall.
B water moves through the cell surface membrane.
C water moves through cytoplasm and tonoplast of vacuoles.
D water moves through the cytoplasm and plasmodesmata.

29 Cool temperatures in natural region 1 of Zimbabwe reduce the incidence of

A virus and fungi.
B viruses transmitted by aphids.
C fungal diseases spread by air.
D bacterial diseases transmitted by air.

30 Which factor is true about anion exchange?

- **A** increases with valency of anion but also increases with increasing valency of cations
- **B** decreases with decrease in pH
- **C** decreases with increasing salt concentration.
- **D** is much less in 1:1 clays than in 2:1 clays

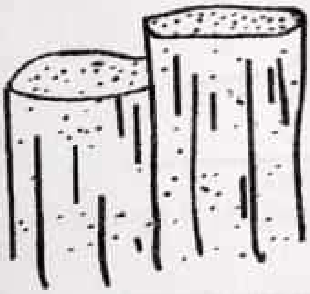
31 Which set of nutrients is composed of basic cations?

- **A** Ca^{2+}, Mg^{2+}, H^{+}.
- **B** Al^{3+}, Ca^{2+}, Na^{+}.
- **C** Al^{3+}, H^{+}, K^{+}.
- **D** Mg^{2+}, Na^{+}, K^{+}.

32 Crassulacean Acid Metabolism (CAM) plants are different from C_3 and C_4 because of

- **A** phosphoenolpyruvate and malic acid pathways.
- **B** malic acid and ribulose bisphosphate carboxylase pathways.
- **C** Ribulose bisphosphate carboxylase and phosphoenolpyruvate pathways.
- **D** phosphoenolpyruvate pathway only.

33 The following diagram shows a structural type of soil recognised in a soil profile.



Identify the structure.

- **A** platy
- **B** columnar
- **C** prismatic
- **D** granular

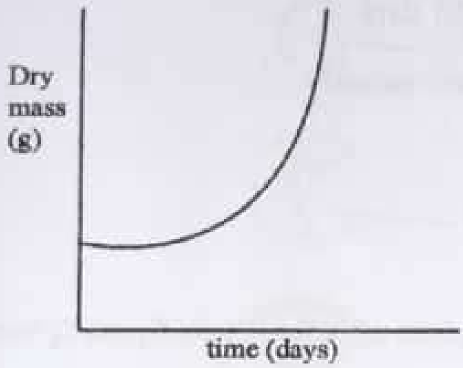
34 A core soil sample which weighs **650g** fits in a **500cm³** cylinder. What is the bulk density of the sample?

- **A** 2.6g/cm³
- **B** 2.6g/m³
- **C** 1.3g/cm³
- **D** 1.3g/m³

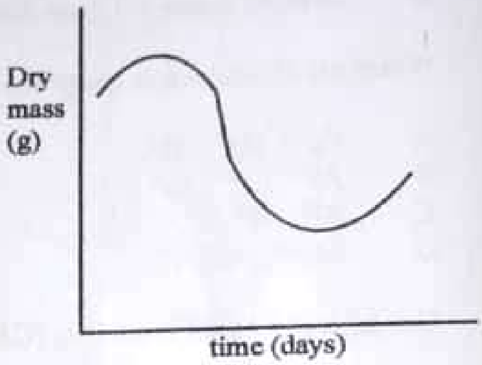
1,3g/cm³ mass $\frac{650}{500}$

Scanned by CamScanner

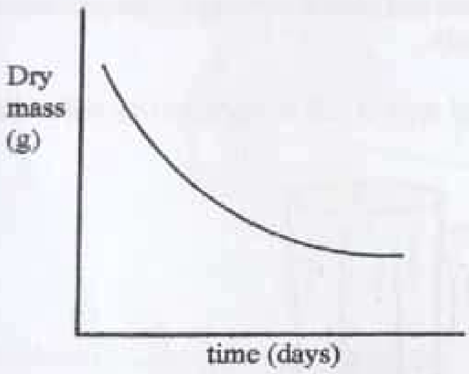
35 An ~~expenditure~~ experiment was conducted to measure the changes in the dry matter content of germinating seedlings up to the onset of photosynthesis.



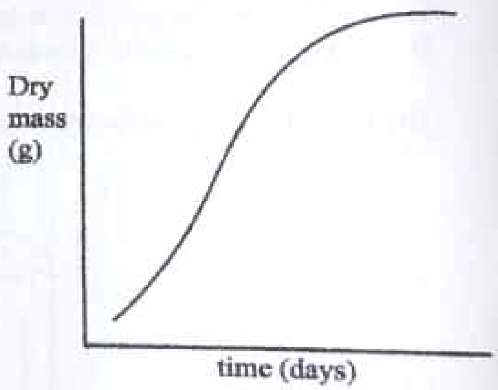
A



B



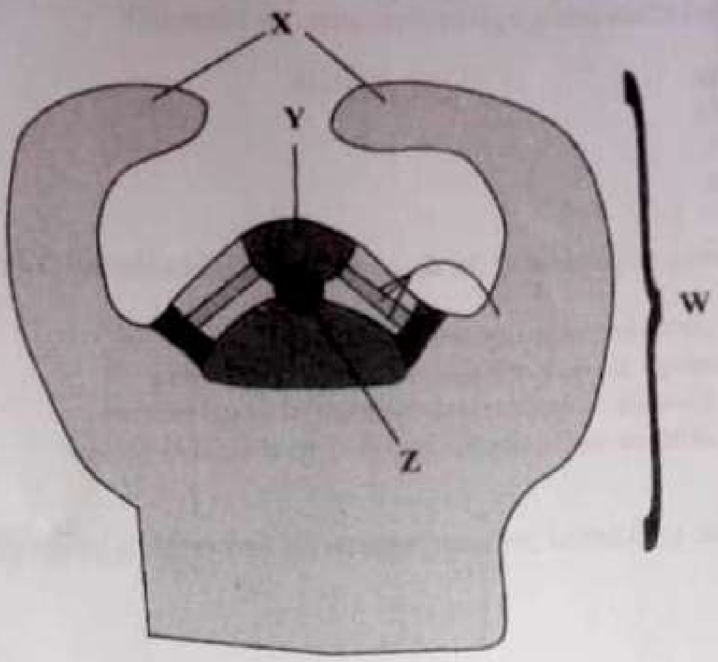
C



D

Which graph correctly potrays the changes.

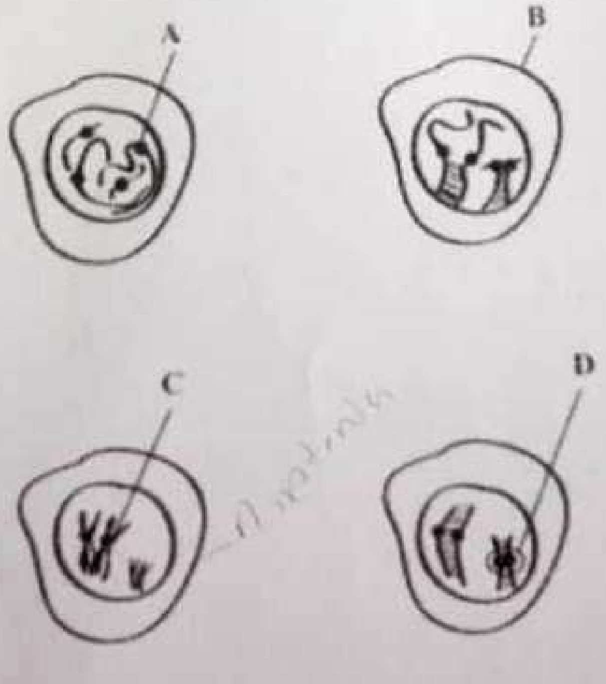
36 The diagram shows an apical meristem.



Which letter represents the leaf primordia?

A W
B X
C Y
D Z

37 Which diagram represents the pachytene stage in prophase 1 of meiosis 1?



38 A farmer planted soya beans at a spacing of **75**cm × **5**cm. Only one seed was placed at each planting station. Calculate the plant population per hactare.

- **A** 266 666
- **B** 375 000
- **C** 375 111
- **D** 26 666

39 Why is the energy requirements for C_4 photosynthesis double that for C_3 photosynthesis?

- **A** carbon dioxide molecules are also doubled
- **B** every carbon dioxide molecule has to be fixed twice
- **C** carbon dioxide molecules and pyruvate have to be fixed
- **D** carbon dioxide molecules have to pass through two phases

40 What is the plant population per hectare when the row width is **80**cm and inter-row **30**cm?

- **A** 44 444
- **B** 41 444
- **C** 44 404
- **D** 41 440

$= \frac{10\,000}{(75 \div 100) \times (5 \div 100)}$

1m = 100cm

$\frac{10\,000}{(40 \div 100) \times (3 \div 100)}$